

Grandpa's **TIME MACHINE**

Sadly, grandpa Rick has been sent home from his nursing home. Some say it's due to bad behaviour towards his nurse ... Anyways, now he lives with you, but that shouldn't mean you can't have any fun together! Despite your quite confused grandpa still not understanding that the year isn't 3013 and that time machines still do not exist - you want to make a program for him that emulates a time machine! Well, at least a text based in Java ... (He can't see the difference anyhow).

GRANDPA'S TECHNICAL DESCRIPTION

(N.B.: Looks like he is missing quite some formal training in presenting his needs ...)

```
#ffaa04-log:01 //
- It should be able to do time machine stuff...
- Remember to buy toothpaste.
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```

As grandpa isn't that good at describing his needs (in text) you allow yourself to ask him an additional time - this is what he told you:

GRANDPA'S WISHES

1) *"If just this time machine could allow me to move back and forward in time by using the same method taking a destination year as a parameter! (And then tell me which year I end up in when I forget.) It could also be super interesting if it could tell me whenever a year went by while travelling and telling me when I reach my destination!"*

2) *"Additionally, I am a true (self-made) historian - you know that! So, if it could also tell me when I pass a year with an important event!"*

3) *"It should also tell me whenever it is a leap year, because I love that stuff! And it must be very precise about this - also for the years when you skip leap years!"*

Additionally, grandpa has given a visual sketch of what he would want the machine to print for him when he travels from the year 2016 to 2019:

```
Tim3M4chin3: Tim3M4chin3: current year is now 2016
Tim3M4chin3: A leap year just happend WoOoOOoOW!
Tim3M4chin3: Current year is now 2017
Tim3M4chin3: Current year is now 2018
Tim3M4chin3: A lot of awesome people went to BootIT
Tim3M4chin3: Current year is now 2019
Tim3M4chin3: You arrived to your destination: 2020
```

GRANDPA'S HINTS AS AN TI-99/4A EXPERT

He should have kept to Texas Instruments!

- 1) It can be useful to think of the time machine as an object; in this way you can store the current year as a field and access it in two methods: one for travelling forward and one for travelling backwards in time.
 - a) When you travel in time it is important to track whether you travel forward or backwards in time when you create your loop! (If you are smart about it, you can manage with a single loop - otherwise, try with two different ones; one for travelling each way)
 - b) Remember to print every year.
- 2) It is not so easy to calculate leap years, so to do so - look at [wikipedia](https://en.wikipedia.org/wiki/Leap_year).

GRANDPA'S DREAM MACHINE

And it's not the new improved Bosch washer

- 1) Add more historical events!
- 2) Change the visual expression so that you can see if multiple messages are from the same year, or that you only print the most important message from the year (e.g., year change < leap year < historical event)
- 3) Grandpa would also love to be able to toggle messages for both year change and leap year (*Hint: use boolean fields and if-statements!*)